

Zachary Heylmun

Principal Engineer

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Professional Experience

 **Principal Engineer @Luminar Technologies**
⌚ 2018 - Present 📍 Orlando, FL

- Sensor interface design and SDK owner for Luminar Technologies Lidar products
- Lidar Firmware architect
- Embedded DSP and control system development for real-time signal processing in sensor firmware
- Developed patented embedded algorithms for range disambiguation in Lidar systems
- Developed OpenCL compute pipeline for early fusion of Lidar and camera data
- Built a custom OpenGL rendering engine optimized for real-time visualization of large-scale point clouds for sensor engineering
- Company owner for software development processes to meet ASPICE and ISO 26262 standards

 **Chief Engineer @Synaptic Sparks**
⌚ 2014 - 2022 📍 Orlando, FL

- Developed app development curriculum for high school students
- Developed open source CNC assembly and usage curriculum for high school students
- Contributed extensively to the Army's open source GIFT (Generalized Intelligent Framework for Tutoring) program
- Developed enrichment technologies for Nemours Children's Hospitals
- Donated STEM teambuilding experiences to students through Digital Escape Velocity

 **Chief Scientist @Digital Escape Velocity**
⌚ 2016 - 2020 📍 Orlando, FL

- Developed VR rendering solutions for immersive escape room experiences
- Built a modular escape room platform with VR, touch, and physical interfaces supporting diverse experiences in the same room
- Utilized IoT technologies to enhance the user experience, eliminating elements which detract from the immersion

 **Owner, CEO @Voidstar Solutions**
⌚ 2013 - Present (Inactive) 📍 Orlando, FL

- Successful deliveries in a variety of domains: financial data, virtual real estate, social networking, and IoT hardware
- Developed web applications, embedded systems, backend services, and analytics, alongside mobile and desktop software
- In 6 years of active operation, every single development contract was extended or expanded upon delivery
- Ceased active operations in 2019 after accepting a full time role at Luminar Technologies as Embedded Systems Architect

 **Software Engineer @Lockheed Martin**
⌚ 2011 - 2013 📍 Orlando, FL

- Built a DirectX 11 rendering engine for Lockheed's Prepar3D flight simulator from scratch on a team of 3
- Collaborated on a novel WGS84 -> matrix compute pass enabling world-scale draw distances without z-fighting or loss of precision
- Assisted in the design and development of a novel rendering pipeline which could render the entire globe at high resolution with no loading screens
- Worked extensively with advanced virtual and augmented reality technologies and optics
- Developed a CUDA-based optics solver to support the development of advanced augmented reality display systems
- Developed Windows drivers for high-speed transfer of flight sim images to FPGA based headset display

Education



Bachelor of Science in Digital Arts and Science Engineering (Computer Graphics), Magna Cum Laude @University of

Florida

⌚ 11/2010 📍 Gainesville, FL

Engineering Expertise



Software Engineering @experience

- GPU compute and graphics: CUDA, OpenCL, DirectX 11, custom rendering engines, shader development
- Embedded systems and DSP: real-time signal processing, firmware architecture, FPGA interfaces
- Strong background in requirements driven design and development of high-throughput firmware and software systems
- Developed systems to a variety of quality and safety standards: CMMI, SPICE, ASPICE, ISO26262, MISRA, DO160
- Delivered production software in Rust, C, C++, CUDA, C#, Dart, Java, Kotlin, Swift, Objective-C



Open Source Contributions and Personal Projects @OSS

- Active contributor to Rust OSS, including the embedded ecosystem, drivers, RSA, Rust Analyzer
- Hardware: Open source hardware to improve safety in challenging technical diving environments
- CNC/3D Printing: Designed and built a variety of 3D printers and CNC routers for the escape room and other fabrication projects